

Sanjay Keerthi

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Professional Summary

Data Scientist with 2+ years of combined industry and research experience designing, auditing, and validating ML systems across deep learning, NLP, and time-series domains. Experienced in identifying model failure modes — including overfitting, data leakage, and imbalanced dataset handling — and translating findings into rigorous, structured technical documentation for specialist and general audiences. MSc Data Science candidate at the University of Manchester (Reinforcement Learning, Statistics & Machine Learning, Data Engineering). Proficient in Python, PyTorch, TensorFlow, Scikit-learn, and SQL; experienced in cross-validation strategies, dimensionality reduction, and Bayesian inference; hands-on with cloud deployment on AWS and Snowflake.

Technical Skills

Programming Languages	Python, SQL, R, Java, C
Machine Learning / AI	Supervised & Unsupervised Learning, Deep Learning, NLP, Computer Vision, Time-Series Forecasting, Reinforcement Learning, Bayesian Inference, Dimensionality Reduction (PCA, t-SNE), Cross-Validation Strategies, Imbalanced Dataset Handling, Overfitting Detection & Correction, Data Leakage Identification, Hyperparameter Optimisation, Model Validation, Model Deployment, AI Evaluation
Libraries & Frameworks	PyTorch, TensorFlow, Scikit-learn, Keras, NumPy, Pandas, Statsmodels, Matplotlib, Seaborn, SciPy
Tools & Platforms	Git, Docker, AWS, MLflow, Airflow, FastAPI, Streamlit, Tableau, Hadoop, Snowflake, Hugging Face, Ollama, Excel
Data Practices	Data Quality Assurance, AI-Generated Code Auditing, Ground-Truth Solution Development, Data Privacy & Security, Unstructured Data Processing, Technical Documentation, Cross-functional Collaboration

Education

The University of Manchester

Sep 2025 – Sep 2026

MSc Data Science Manchester, UK

- Completing core modules in Statistics & Machine Learning, Reinforcement Learning, and Data Engineering Technologies, building rigorous foundations in cross-validation strategies, probabilistic modelling, and scalable data pipelines.
- Partnered with a team of 5 to apply advanced ML methods to real-world datasets, improving model interpretability and reducing validation error by 18% through systematic hyperparameter optimisation and feature engineering.
- Conducting an independent research project applying reinforcement learning to dynamic optimisation problems; managing project timelines autonomously in a hybrid remote environment, bridging academic research with industry-grade implementation.

Panimalar Engineering College

Aug 2021 – May 2025

B.E. Computer Science — CGPA: 8.58/10 — Merit Scholarship (INR 25,000) Chennai, India

- Applied supervised and unsupervised learning techniques to real-world problems, including Traffic Congestion Prediction (LSTM & Attention Mechanism) and California Housing Price Forecasting on large-scale structured datasets.
- Engineered end-to-end data pipelines — preprocessing, dimensionality reduction, feature engineering, and cross-validation — improving model accuracy by 22% and reducing data preparation time by 40%.

Professional Experience

Data Scientist Intern

Jan 2025 – Jul 2025

Vcodez — Remote

- Engineered LSTM and ARIMA time-series forecasting models for real-time climate monitoring on large-scale structured datasets (10M+ rows), achieving **91% accuracy** and enabling data-driven decisions for cross-functional stakeholders.
- Deployed a Streamlit dashboard integrating live API feeds, lifting user engagement by **35%** through animated visualisations and a responsive UI; produced end-to-end technical documentation for both engineering and non-technical audiences.
- Automated end-to-end data ingestion, preprocessing, and quality validation pipelines — enforcing data privacy and security standards throughout — eliminating **6+ hours of manual work per sprint. weekly status reporting and knowledge transfer documentation.**
- Audited and validated deep learning research implementations in PyTorch and TensorFlow for technical correctness and efficiency, translating findings into structured feedback that improved production model performance; identified and corrected overfitting and data leakage issues across multiple model iterations.

Data Analytics Intern

PrepInsta — Remote

Jun 2024 – Oct 2024

- Applied Python and Tableau to conduct statistical analysis, data mining, and EDA on large-scale structured and unstructured datasets, cutting reporting turnaround time by **25%** and delivering precisely documented insights to technical and non-technical audiences.
- Built a Tableau dashboard tracking 2024 Indian General Election campaign spending across **500+ constituencies**, surfacing funding patterns and potential voter influence zones through SQL-based aggregation and visualisation.
- Partnered with a 4-member cross-functional analytics team to develop and tune classification models, improving F1-score by **15%** through supervised learning techniques and feature engineering.

Research Experience

Research Assistant

Apr 2024 – Jul 2024

Panimalar Engineering College — Supervisor: Dr. Santhanalakshmi S. T. Chennai, India

Custom Deep CNN with YOLOv3 Backbone for Multiple Object Detection

- Developed a GAN-based image dehazing module integrated into a YOLOv3 backbone, improving object detection accuracy by **19%** in foggy and low-visibility environments and contributing labelled outputs to an AI training dataset pipeline.
- Reduced false-positive detections by **23%** across diverse unstructured test scenarios by optimising the dehazing-YOLOv3 integration, validated through rigorous cross-condition performance testing and structured error analysis.
- Documented ground-truth experimental methodologies and model debugging results for 10–20 validation experiments, producing structured reference solutions that supported PhD thesis preparation.

Projects

Traffic Flow Optimisation Using LSTM

Python, PyTorch, Streamlit

- Built a sequence-to-sequence LSTM forecasting hourly traffic volumes on the Minneapolis–St. Paul corridor across 1–2 years of historical data, achieving **RMSE of 12.4** using historical, weather, and temporal features extracted from large-scale structured datasets.
- Reduced forecast error by **21% over the ARIMA baseline** through lag feature engineering, input normalisation, and hyperparameter optimisation — diagnosing and correcting overfitting across multiple cross-validation folds.
- Integrated the trained model into a Streamlit dashboard, reducing average route-planning decision time by 3 hours weekly; documented the methodology end-to-end as a structured technical reference for stakeholder review.

Number Plate Detection

Python, YOLOv3, OpenCV, OCR

- Achieved **94% precision and 92% recall** on test video streams by implementing an end-to-end automatic number plate detection system using a YOLOv3-based CNN with OCR text extraction, validated across diverse unstructured video data.
- Reduced detection failure rate by **30%** under poor lighting and oblique-angle conditions by applying contrast normalisation and edge enhancement preprocessing — identifying and correcting model failure modes through structured performance testing.

Certifications

- Machine Learning & Deep Learning Specialisation — Coursera / DeepLearning.AI (2024)
- GitHub Foundation — GitHub (2024–2027)
- Oracle Certified Foundation Associate — Oracle (2024–2026)
- AI for Real World Applications — TCS (2023)

Relevant Coursework

Probability, Statistics & Stochastic Processes • Pattern Recognition & Machine Learning • Data Engineering Technologies • Reinforcement Learning • Linear Algebra • Differential Equations

Activities & Leadership

- **Smart India Hackathon (SIH) & Police Hackathon** — Participant, Panimalar Engineering College
- **AWS Student Club** — Member; organised cloud-focused workshops and hands-on labs for 50+ students
- **Coder Forum Club** — Member; regular contributor to coding challenges and peer-led study sessions
- **Open Source** — Active contributor to applied ML projects and open-source development on GitHub